MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG5DSCBOT301** Course Title: **Plant Cell and Molecular Biology**

Time: 1.5 Hrs Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create
Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A: Very Short Answer Questions
Answer 10 questions
Each question carries 1 mark

	Questions	Taxonomy	CO
1.	Name any two evolutionarily conserved proteins present in plants.	Remember	5
2.	What are split genes?	Remember	6
3.	Who proposed the term “Cell”?	Remember	1
4.	Mention any two functions of Lysosome.	Remember	2
5.	Explain the major functions of SER.	Understand	2
6.	Comment on the significance of DNA repair in cells.	Evaluate	6
7.	What are the major components in cytoskeleton?	Understand	2
8.	What are Dictyosomes?	Remember	2
9.	Explain missense mutations.	Evaluate	6
10.	What is telomere?	Understand	3
11.	What are the functions of nucleolus?	Understand	3
12.	Mention any two characteristic features of Z DNA.	Understand	5

Part B: Short answer Questions
Answer 6 questions
Each question carries 5 marks

13.	Write a comparative account on different types of RNA.	Analyze	5
14.	Describe the organization of DNA into chromosomes.	Understand	3
15.	Evaluate the roles of major enzymes involved in prokaryotic DNA replication.	Evaluate	6
16.	Compare the structural and functional characteristics of heterochromatin and euchromatin	Understand	3
17.	Describe the structural organization of plasma membrane.	Understand	2
18.	Analyze the molecular mechanisms involved in apoptosis.	Analyze	5
19.	Justify the significance of meiosis in the plant kingdom.	Analyze	5
20.	Explain the mechanism of gene regulation in prokaryotes by operon model, using a suitable example.	Evaluate	4

Part C: Essay Questions
Answer 1 question
Each question carries 10 marks

21.	Describe the structure and major functions of semi-autonomous organelles, and explain why they are referred as "semi-autonomous" organelles.	Understand	2
22.	Describe the molecular mechanism of translation in prokaryotes.	Understand	6